# Baby on board

# 14a

When a baby travels in a car, the baby must be in a specially designed baby or child restraint. That restraint helps to protect the baby by keeping him or her in place as the car changes speed and/or direction. It also provides maximum protection if there is an accident. Imagine that a car is completely full, with all seats occupied. Someone does an illegal, dangerous and stupid thing, taking the 6 kg baby to hold on their lap. There is an accident, with the car hitting a brick wall at a speed of 20 km/h.

We will calculate what force is needed to keep the baby on the holder's lap when a car comes to a sudden stop. Newton's second law of motion can be used here. Newton found that multiplying the mass of an object (*m*, measured in kg) and its acceleration (*a*, measured in m/s/s) gives the force needed to bring the object to rest (*F*, measured in N).

**1** Write Newton's force equation. _____

Assume that a car takes a quarter of a second to come to a complete stop after hitting a wall. (As soon as the car hits the wall, the bumper bar and the metal begin to crumple and it takes some time for the car to be at a total standstill.)

**2** Calculate the change in speed of the car. _____

Note that the negative change in speed shows that the speed is decreasing.

**3** Convert the speed change in km/h to a speed change with the unit m/s. (Hint: Divide the km/h speed by 3.6 to express it in m/s.)

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**4** Write the formula for calculating acceleration.
(Hint: Acceleration is defined as the change of velocity in a unit time.) _____

**5** Calculate the acceleration of the car (and anybody wearing a seat belt in the car!). The unit of acceleration will be m/s/s (or $m/s^2$, as it is most often written).

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Note that the acceleration is negative. Because it is a negative acceleration, it is called a deceleration. The negative sign indicates that acceleration slows the car. The negative sign is often dropped in calculations and remembered as the direction of the acceleration.

**6** Substitute the value for the acceleration in Newton's force equation to find the force that is needed to keep hold of the 6 kg baby.

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**7** If this force was a weight and pulled downwards (instead of horizontally, as it would if the baby were on a person's lap), we can calculate an equivalent mass that would experience the same force. In other words, we can equate the deceleration force on the baby with the mass of an imaginary baby being held up against the force of gravity.

**a** Write Newton's force formula again. _____

**b** Rearrange the formula so that mass is its subject. _____

**c** The size of the force is known from Question 6. Acceleration due to gravity is 10 $m/s^2$.
Substitute these values into the mass formula and calculate the mass of the imaginary baby being held up against the force of gravity.

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We have just calculated how big the deceleration force on a 6 kg baby appears to be when someone is trying to keep the baby on a lap during a 20 km/h crash. In other words, the baby seems to have a mass of the answer in 7c, when his or her actual mass is only 6 kg.


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**14b**

**8** Use your new understanding from this activity to find how much mass a 6 kg baby would appear to have if a car travelling at 50 km/h crashed and was brought to a stop within a quarter of a second. Show your working.

**9** Find how much mass the baby would appear to have if the car was travelling at 80 km/h, crashed and was bought to a stop in a quarter of a second. Show your working out.

**10** A different baby has a mass of 10 kg, and was in a car travelling at 110 km/h. The car crashed, coming to a standstill in 0.25 s. What was the deceleration force on the baby? Show your working out.

**11** Two passengers in a car chose not to wear their seatbelts. One passenger has a mass of 50 kg and the other passenger has a mass of 70 kg. The car was travelling at 95 km/h when it crashed, coming to a standstill in 0.5 s. What was the deceleration force on each passenger? Show your working out.

**12** Imagine a family car (mass of the car plus occupants is 1500 kg) coming to a quick stop (in 0.5 s) from a speed of 150 km/h (some roads in the Northern Territory have no speed limit). Calculate the deceleration force for the whole car. Show your working out.

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